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Morgan Castle

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DSC 623 - Project - Conceptual Design

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A. Main Entities:
    a. Outlet
        i. The outlet represents a physical location of the company where vehicles are stored and staff members work. The system must keep track of each outlet in order to track their vehicles and employee locations.
    b. Client
        i. Clients are the individuals who rent the vehicles. The system must maintain records of the clients so that the hire agreements are associated with the right individuals.
    c. Vehicle
        i. Vehicles are the primary asset of the company. Each vehicle is uniquely identified via its registration number, allowing the system to store and monitor important information about each vehicle.
    d. Staff
        i. The staff represent the employees that work for the company. The system needs to store information about each employee and identify the outlet they work in.
    e. Hire Agreement
        i. Each Client signs a hiring agreement with the Company upon renting a vehicle. This allows the Company to keep track of who has one of their vehicles at any given time.

B. Entity Relationship(s)
    a. Outlet - stores - Vehicle
    b. Outlet - employs - Staff
    c. Client - signs - Hire Agreement
    d. Vehicle - assigned - Hire Agreement

C. Multiplicity Constraints
    a. Outlet to Vehicle = 1:*
        i. An outlet can store many vehicles
        ii. Each vehicle is stored at one outlet at a time
    b. Outlet to Staff = 1:*
        i. An outlet can employ many staff members
        ii. An employee can only work at one outlet
    c. Client to Hire Agreement = 1:*
        i. A client can sign many hire agreements
        ii. Each hire agreement belongs to one client
    d. Hire Agreement to Vehicle = *:1
        i. A vehicle can appear in many hire agreements

ii. Each hire agreement agreement involves one vehicle

Everything that was explained above is represented visually in this chart

E1	M1	Relation	M2	E2	C
Outlet	1..1	Stores	1..*	Vehicle	1..*
Outlet	1..1	Employs	1..*	Staff	1..*
Client	1..1	Signs	1..*	Hire Agreement	1..*
Vehicle	1..1	Assigned	1..*	Hire Agreement	1..*

D. Entity Attributes

a. Outlet
   i. OutletID (PK)
   ii. Address
   iii. PhoneNumber
   iv. FaxNumber

b. Client
   i. ClientID (PK)
   ii. FirstName
   iii. LastName
   iv. Address
   v. PhoneNumber
   vi. DateOfBirth
   vii. DrivingLicenseNumber

c. Vehicle
   i. RegistrationNumber (PK)
   ii. Model
   iii. Make
   iv. EngineSize
   v. Capacity
   vi. CurrentMileage
   vii. DailyHireRate
   viii. CurrentLocation

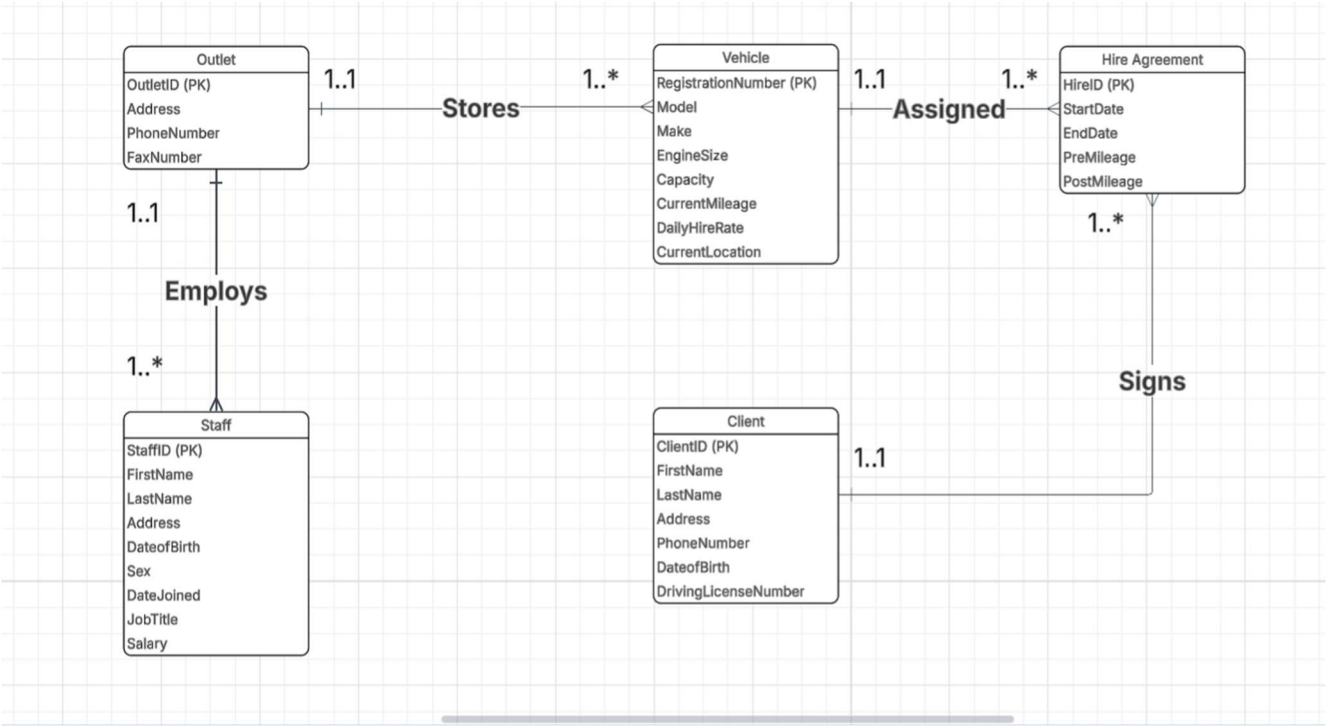
d. Staff
   i. StaffID (PK)
   ii. FirstName

iii. LastName
iv. Address
v. PhoneNumber
vi. DateOfBirth
vii. Sex
viii. DateJoined
ix. JobTitle
x. Salary

e. Hire Agreement
i. HireID (PK)
ii. StartDate
iii. EndDate
iv. PreMileage
v. PostMileage

E. Candidate and Primary Keys
a. Outlet
i. Outlet ID (PK)
ii. Candidate Keys: Outlet ID, address, phone number, Fax Number (assuming each branch has a unique address, phone number, and Fax Number)
b. Client
i. Client ID (PK)
ii. Candidate Keys: Client ID, Driving License Number
c. Vehicle
i. RegistrationNumber (PK)
ii. Candidate Keys: Registration Number
d. Staff
i. StaffID (PK)
ii. Candidate Keys: StaffID
e. Hire Agreement
i. Hire ID (PK)
ii. Candidate Keys: Hire ID

F. E-R Diagram


Outlet
OutletID (PK)
Address
PhoneNumber
FaxNumber
1..1
Stores
1..*
Vehicle
RegistrationNumber (PK)
Model
Make
EngineSize
Capacity
CurrentMileage
DailyHireRate
CurrentLocation
1..1
Assigned
1..*
Hire Agreement
HireID (PK)
StartDate
EndDate
PreMileage
PostMileage
1..1
Employs
1..*
1..*
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1..1